

Problem Statement No. 8 – Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with diet planning, portion control, and tracking nutritional intake. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without personalized, real-time guidance, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an AI-powered Nutrition Agent that delivers personalized dietary insights and guidance for individuals, families, and health professionals.

Personalized Diet Plans – Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.

Food Data RAG Retrieval – Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.

Diet Tracking & Feedback – Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.

Preventive Health Focus – Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).

Multi-Agent System – Nutrition Knowledge Agent, Diet Recommendation Agent, Health Advisory Agent, and Food Log & Feedback Agent.

Visualization Dashboard – Show daily/weekly nutrient intake, deficiencies, and progress toward health goals.